IOE AND AI DRIVEN SYSTEMS

LAB OUTPUTS

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EXPERIMENT 4:

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# IoE PROTOCOL PERFORMANCE COMPARISON
np.random.seed(24)
protocols = ["Zigbee", "6LoWPAN", "NB-IoT"]
simulation_time = 120       # seconds
packets_sent = 1200
packet_size = 512           # bytes
# Illustrative baseline values (NOT NS-3 measurements)
parameters = {
    "Zigbee":  {"pdr": 92, "delay": 38, "energy": 15, "throughput": 41},
    "6LoWPAN": {"pdr": 89, "delay": 62, "energy": 21, "throughput": 34},
    "NB-IoT":  {"pdr": 97, "delay": 72, "energy": 29, "throughput": 45}
}
results = []
for protocol in protocols:
    p = parameters[protocol]
  # Illustrative variations for demonstration only
```

```
    pdr = np.clip(p["pdr"] + np.random.normal(0, 1.0), 0, 100)
    delay = max(0.1, p["delay"] + np.random.normal(0, 2.5))
    energy = max(0.1, p["energy"] + np.random.normal(0, 0.8))
    throughput = max(0.1, p["throughput"] + np.random.normal(0, 1.2))
    packets_received = round(packets_sent * pdr / 100)
    results.append([
    protocol,
        packets_sent,
        packets_received,
        round(pdr, 2),
        round(throughput, 2),
        round(delay, 2),
        round(energy, 2)
    ])
df = pd.DataFrame(results, columns=[
    "Protocol",
    "Packets Sent",
    "Packets Received",
    "Packet Delivery Ratio (%)",
    "Throughput (kbps)",
    "Average Delay (ms)",
    "Energy Consumption (J)"
])
# DISPLAY RESULTS
print("=" * 85)
print("IoE COMMUNICATION PROTOCOL PERFORMANCE ANALYSIS")
print("=" * 85)
print("Simulation duration:", simulation_time, "seconds")
print("Packet size:", packet_size, "bytes")
```

```
print("Data status: ILLUSTRATIVE EXAMPLE — NOT NS-3 OUTPUT")
print()
print(df.to_string(index=False))
# PERFORMANCE SUMMARY
best_pdr = df.loc[
    df["Packet Delivery Ratio (%)"].idxmax(), "Protocol"
]
best_throughput = df.loc[
    df["Throughput (kbps)"].idxmax(), "Protocol"
]
lowest_delay = df.loc[
    df["Average Delay (ms)"].idxmin(), "Protocol"
]
lowest_energy = df.loc[
    df["Energy Consumption (J)"].idxmin(), "Protocol"
]
print("\n" + "=" * 55)
print("PERFORMANCE SUMMARY")
print("=" * 55)
print("Highest packet delivery ratio :", best_pdr)
print("Highest throughput            :", best_throughput)
print("Lowest average delay          :", lowest_delay)
print("Lowest energy consumption     :", lowest_energy)
# INDIVIDUAL PERFORMANCE GRAPHS
metrics = [
    ("Throughput (kbps)", "Throughput Comparison", "Throughput (kbps)"),
    ("Average Delay (ms)", "Average Delay Comparison", "Delay (ms)"),
    ("Packet Delivery Ratio (%)", "Packet Delivery Ratio Comparison", "PDR (%)"),
    ("Energy Consumption (J)", "Energy Consumption Comparison", "Energy (J)")
```

```
]
for column, title, ylabel in metrics:
    plt.figure(figsize=(8, 5))
    bars = plt.bar(df["Protocol"], df[column])
  plt.title(title)
    plt.xlabel("Communication Protocol")
    plt.ylabel(ylabel)
    plt.grid(axis="y", alpha=0.3)
  if column == "Packet Delivery Ratio (%)":
        plt.ylim(0, 100)
 plt.bar_label(bars, fmt="%.2f", padding=3)
    plt.tight_layout()
    plt.show()
# COMBINED NORMALIZED COMPARISON GRAPH
normalized = pd.DataFrame()
normalized["Protocol"] = df["Protocol"]
higher_is_better = [
    "Packet Delivery Ratio (%)",
    "Throughput (kbps)"
]
lower_is_better = [
    "Average Delay (ms)",
    "Energy Consumption (J)"
]
for col in higher_is_better:
    min_val, max_val = df[col].min(), df[col].max()
    normalized[col] = (
        (df[col] - min_val) / (max_val - min_val)
        if max_val != min_val else 1.0
```

```
    )
for col in lower_is_better:
    min_val, max_val = df[col].min(), df[col].max()
    normalized[col] = (
        (max_val - df[col]) / (max_val - min_val)
        if max_val != min_val else 1.0
    )
normalized = normalized.rename(columns={
    "Packet Delivery Ratio (%)": "PDR",
    "Throughput (kbps)": "Throughput",
    "Average Delay (ms)": "Delay efficiency",
    "Energy Consumption (J)": "Energy efficiency"
})
normalized.set_index("Protocol").plot(
    kind="bar", figsize=(10, 6), ylim=(0, 1.15)
)
plt.title("Normalized Comparative Performance of IoE Protocols")
plt.xlabel("Communication Protocol")
plt.ylabel("Normalized score (higher is better)")
plt.grid(axis="y", alpha=0.3)
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
# SAVE RESULTS
df.to_csv("/content/IoE_protocol_results.csv", index=False)
normalized.to_csv("/content/IoE_normalized_results.csv", index=False)

print("\nCSV files saved:")
print("/content/IoE_protocol_results.csv")
```

```
print("/content/IoE_normalized_results.csv")
# CONCLUSION FROM ILLUSTRATIVE DATA
print("\n" + "=" * 55)
print("CONCLUSION")
print("=" * 55)
print(f"{best_pdr} has the highest illustrative PDR.")
print(f"{best_throughput} has the highest illustrative throughput.")
print(f"{lowest_delay} has the lowest illustrative delay.")
print(f"{lowest_energy} has the lowest illustrative energy consumption.")
print("\nThese findings apply only to the example values.")
print("Replace them with validated measurements before reporting.")
```

OUTPUT:

```
IoE COMMUNICATION PROTOCOL PERFORMANCE ANALYSIS
Simulation duration: 120 seconds
Packet size: 512 bytes
Data status: ILLUSTRATIVE EXAMPLE — NOT NS-3 OUTPUT

Protocol  Packets Sent  Packets Received  Packet Delivery Ratio (%)  Throughput (kbps)  Average Delay (ms)  Energy Consumption (J)
  Zigbee         1200              1120                     93.33              39.81               36.07              14.75
 6LoWPAN         1200              1055                     87.93              34.35               58.40              21.45
  NB-IoT         1200              1144                     95.37              47.27               72.55              29.54

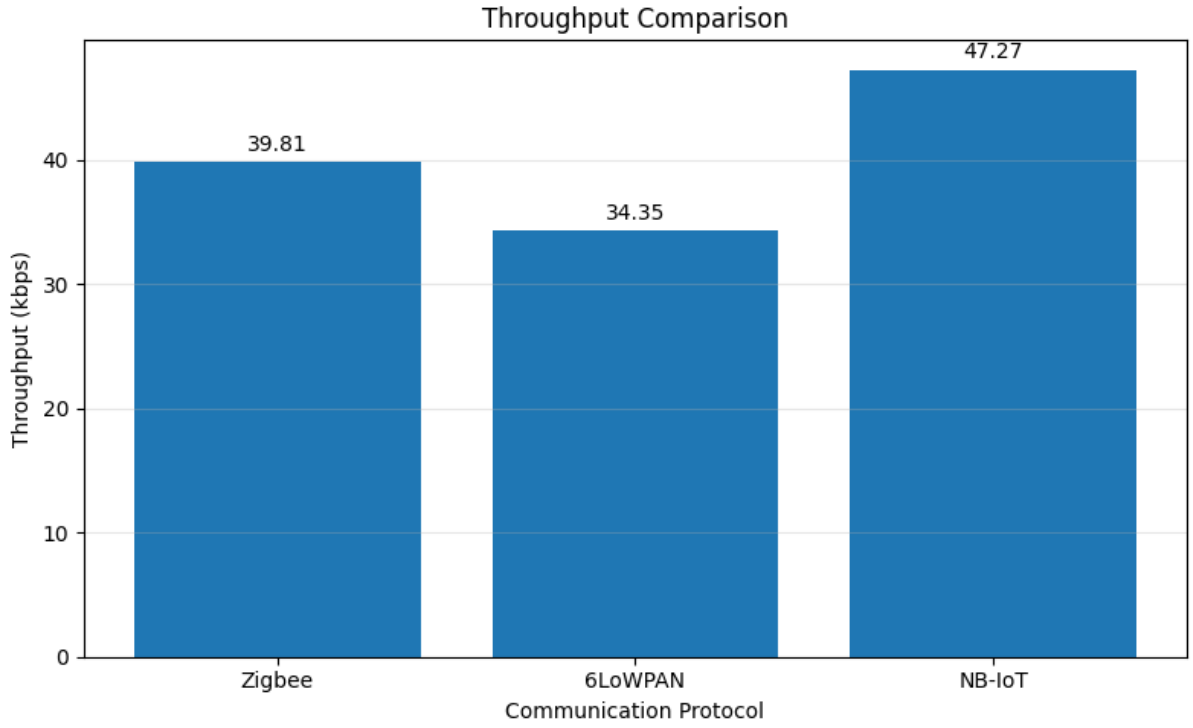
PERFORMANCE SUMMARY

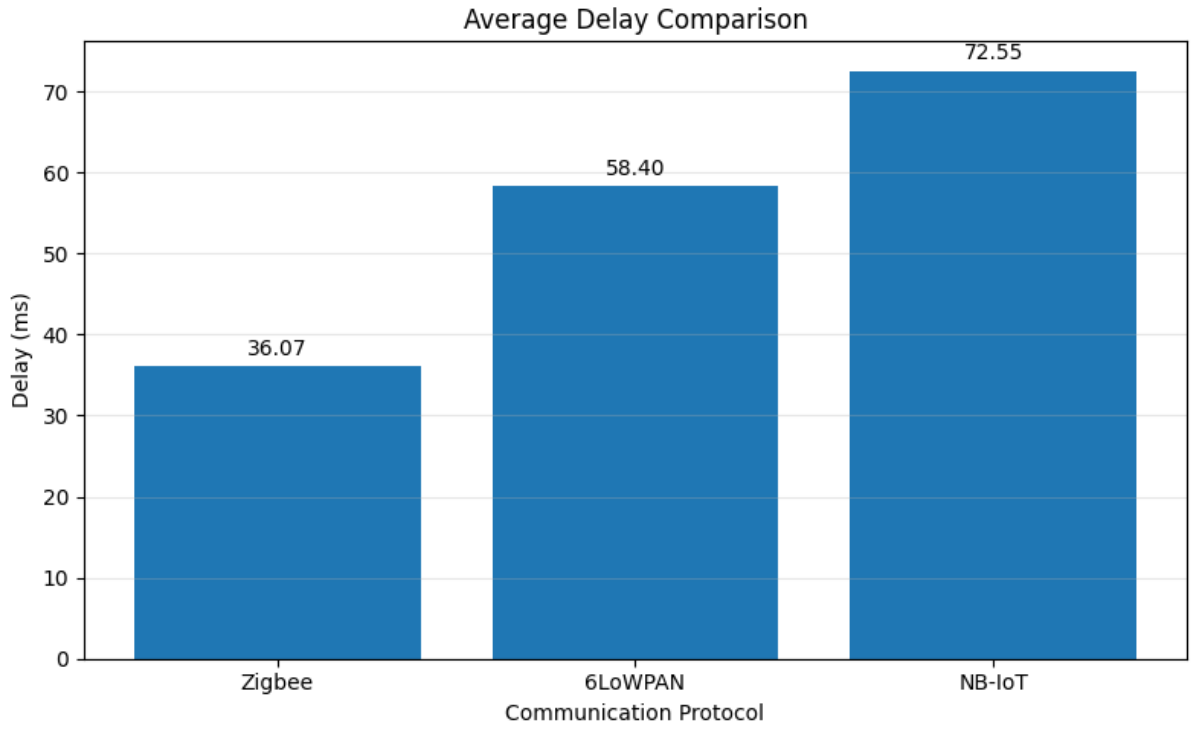
Highest packet delivery ratio : NB-IoT
```

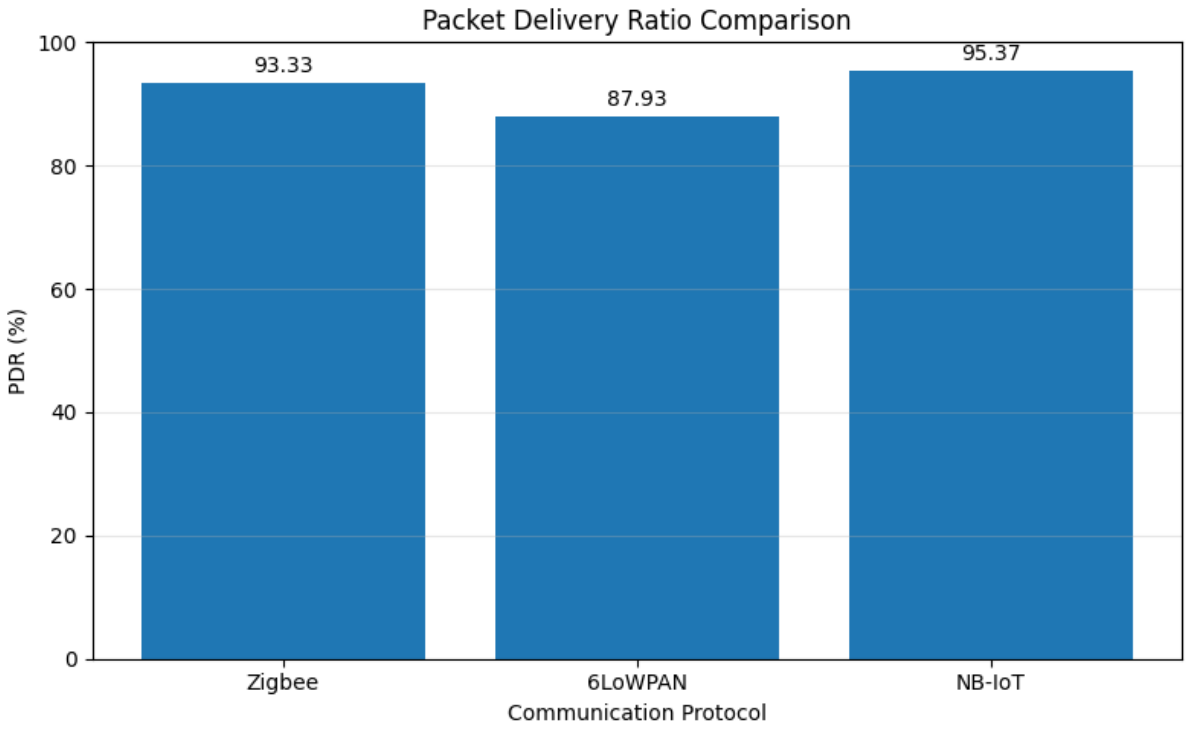
Highest throughput : NB-IoT

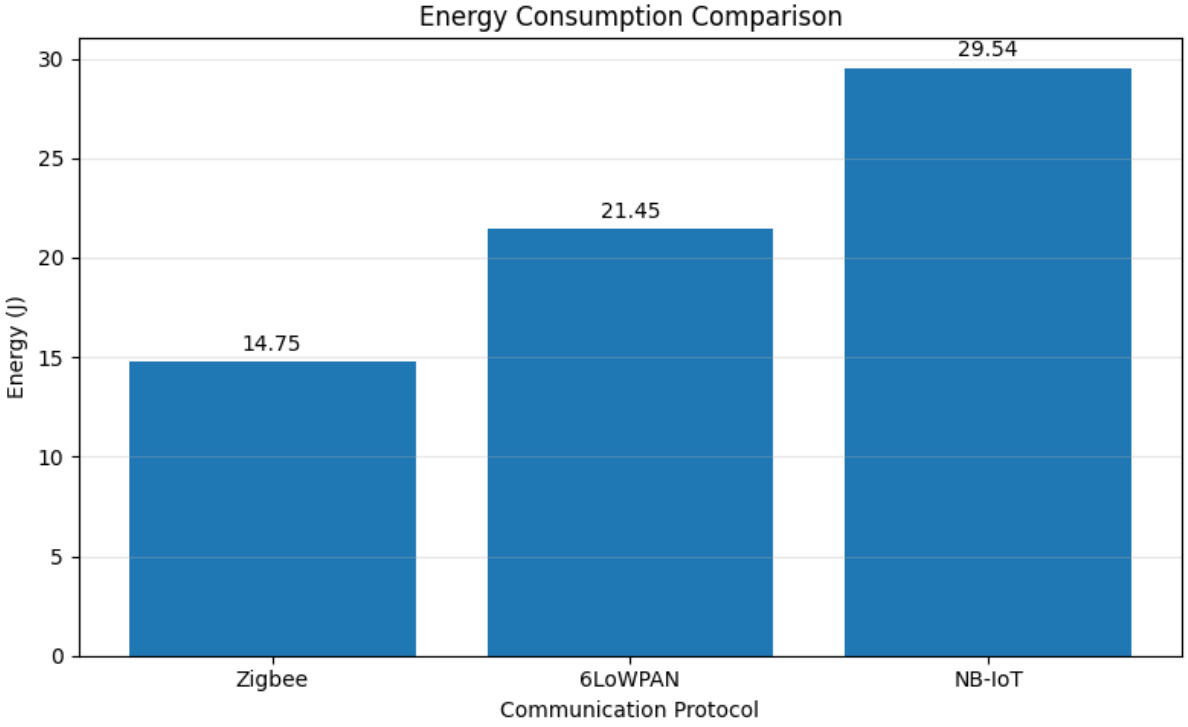
Lowest average delay : Zigbee

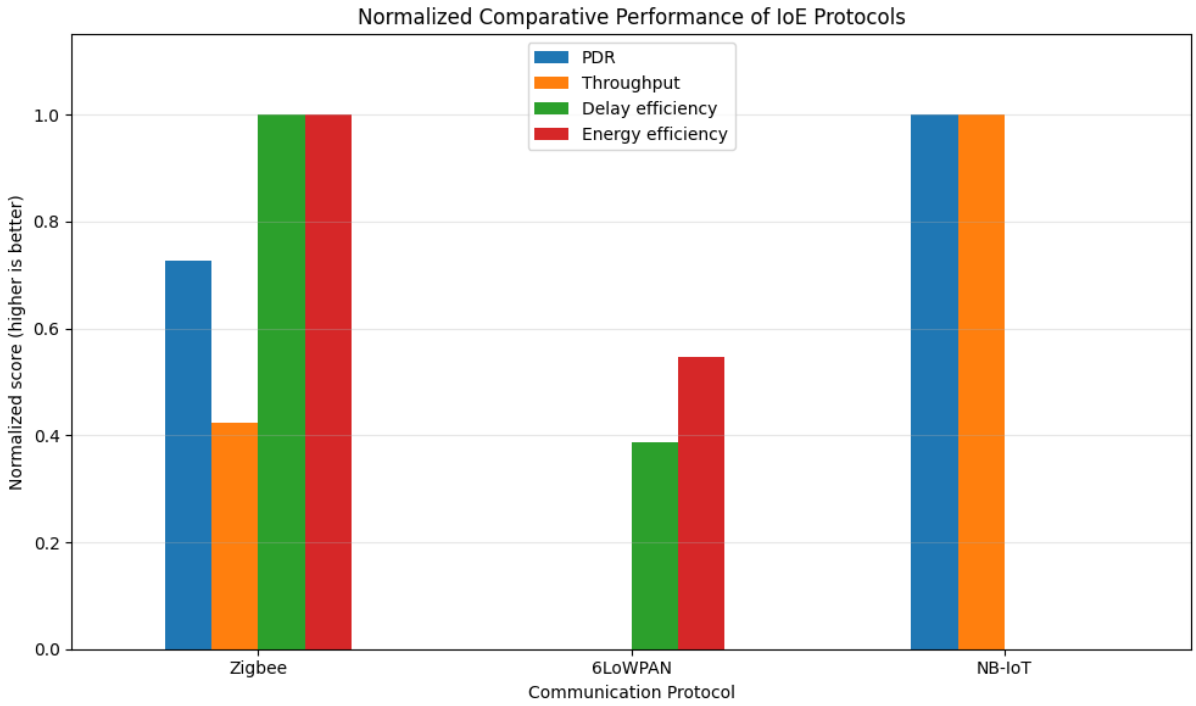
Lowest energy consumption : Zigbee






Packet Delivery Ratio Comparison
PDR (%)
100
80
60
40
20
0
93.33
87.93
95.37
Zigbee
6LoWPAN
NB-IoT
Communication Protocol


Energy Consumption Comparison
Energy (J)
30
25
20
15
10
5
0
14.75
21.45
29.54
Zigbee
6LoWPAN
NB-IoT
Communication Protocol



CSV files saved:

/content/IoE_protocol_results.csv

/content/IoE_normalized_results.csv

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CONCLUSION

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NB-IoT has the highest illustrative PDR.

NB-IoT has the highest illustrative throughput.

Zigbee has the lowest illustrative delay.

Zigbee has the lowest illustrative energy consumption.